commtikz: A TikZ Library for Electro-Optical, Photonics, RF, and Electronic Schematics

Version 0.1.0 (2026/11/10)

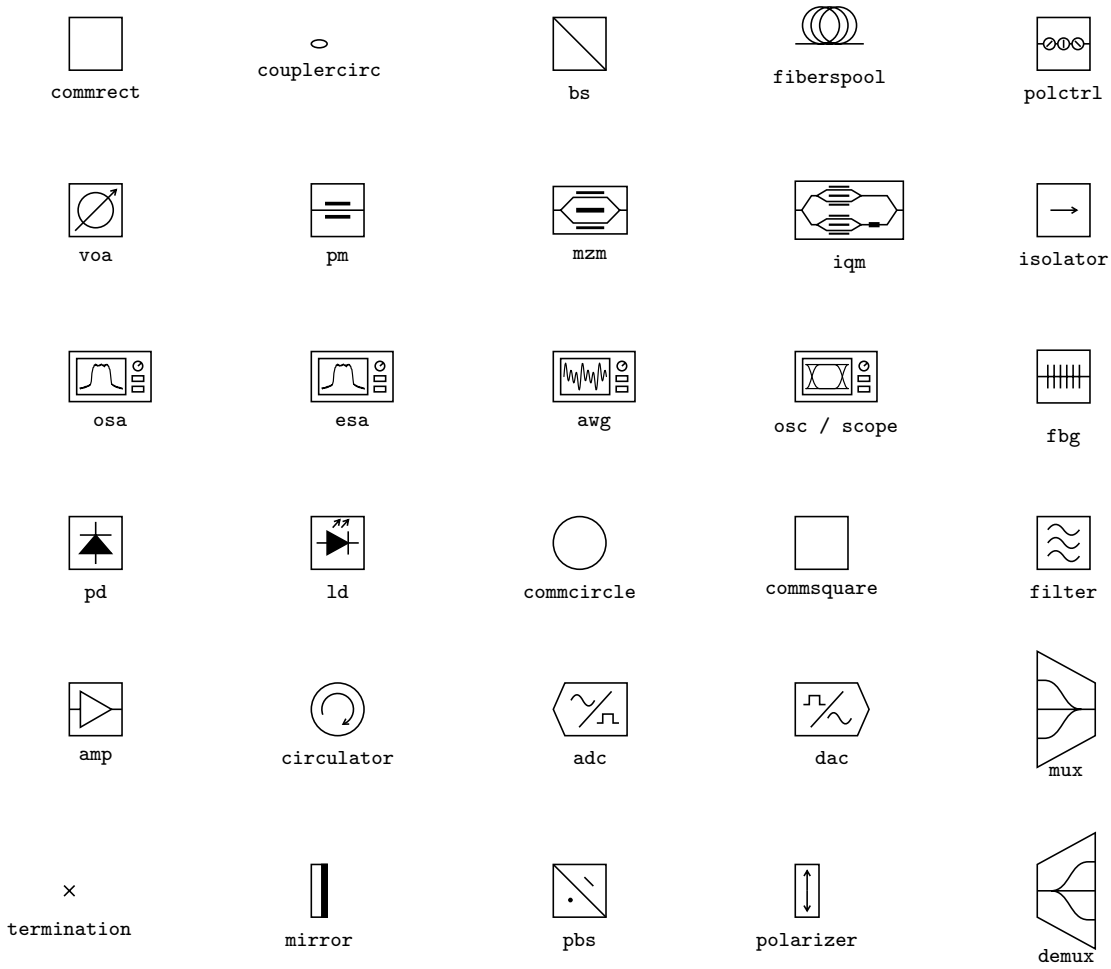
1 Overview & Quick Start

The `commtikz` package provides native PGF/TikZ node shapes, configurable multi-port anchors (`W0..Wn`, `E0..En`, `N0..Nn`, `S0..Sn`), background-layer signal-domain connection styles (`opt`, `rf`, `optrf`, `dc`, `ctrl`), Bezier connectors (`arcz`, `arcn`, `arc2`, `arcloop`), 90°/180° circular turns (`bend`, `opt bend`, `rf bend`, `uturn`), inline `to[...]` placement, and an electronic circuit sublibrary for electro-optical and RF/DSP schematics.

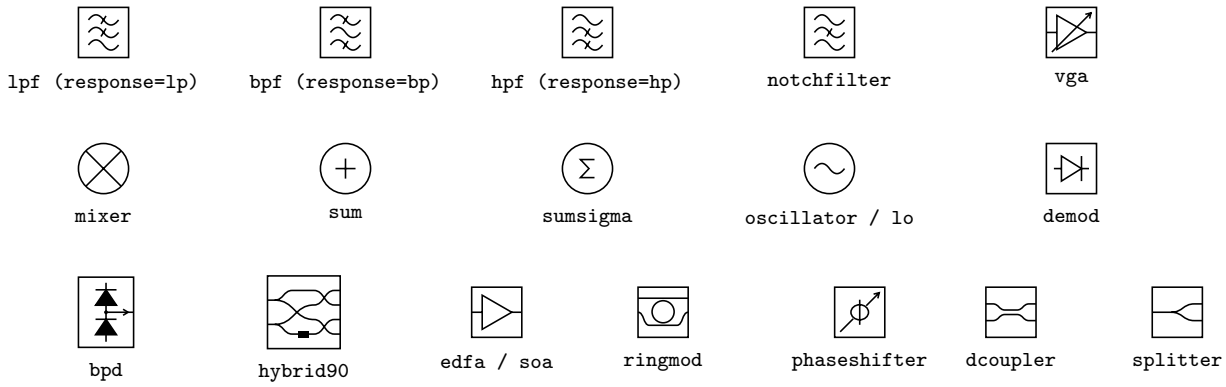
```
\usepackage{commtikz} % or \usepackage{tikz} \usetikzlibrary{commtikz}
```

2 Core Electro-Optical & System Component Grid

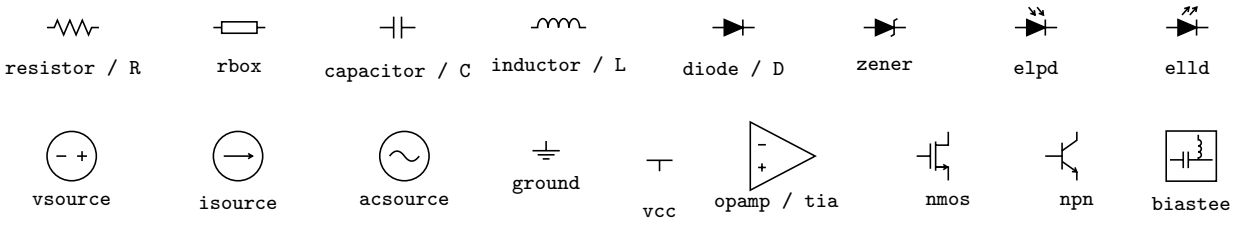
Below is the catalog of the core optical, instrument, and mixed-signal building blocks. Ordinary boxed blocks share a 1×1 footprint, including PM, ringmod, phase shifter, FBG, and the amplifier family. Special aspect ratios are preserved. Defaults use 85% block scale and 110% circuit scale. The oscilloscope screen shows an eye diagram; PC uses outlined paddles with distinct axes.



3 DSP, RF & Coherent Photonics Components



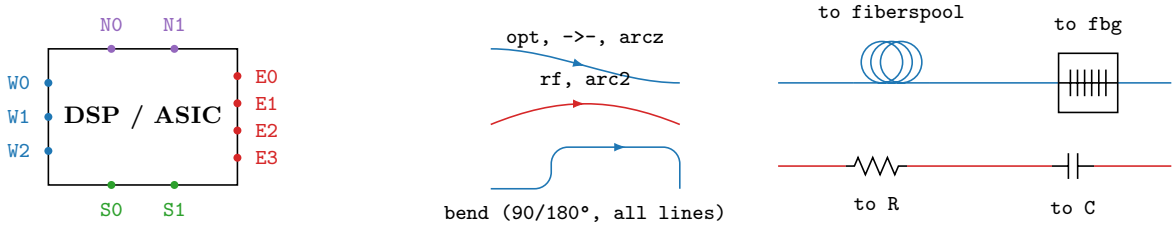
4 Electronic & Electrical Components Sublibrary



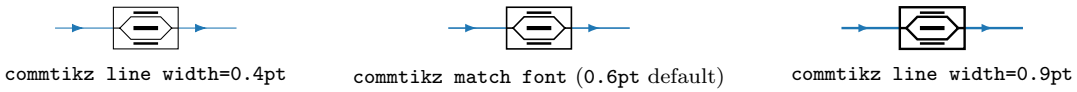
5 Multi-Port Anchor System, Connectors, Scaling & Line Width Control

Generic blocks and mux/demux support `numN`, `numS`, `numE`, `numW` (or `ports={W=2, E=4, N=2, S=1}`), automatically generating evenly spaced anchors `W0..Wn`, `E0..En`, `N0..Nn`, and `S0..Sn` alongside standard TikZ anchors. Fixed-topology blocks enforce their documented W/E counts, top-to-bottom order, and full-height W/E span. Signal connectors (`opt`, `rf`, `optrf`, `dc`, `ctrl`) are drawn on the `background` layer behind all component shapes.

Scaling: `commtikz unit=<dim>` (default `1cm`) or `commtikz scale=<num>` sets the base unit u. Dimensions also use `commtikz block scale` (0.85) or `commtikz circuit scale` (1.10). These factors multiply all dimensionless block keys (`width`, `height`, `height1`, `height2`, `comp radius`) and internal arrows. Pass `commtikz scale=0.75` (or `width=..., height=...`) on a node to scale a single block, or pass `[commtikz unit=0.8cm, x=0.8cm, y=0.8cm]` on `\begin{tikzpicture}` to scale a whole figure while keeping text labels at document font size.

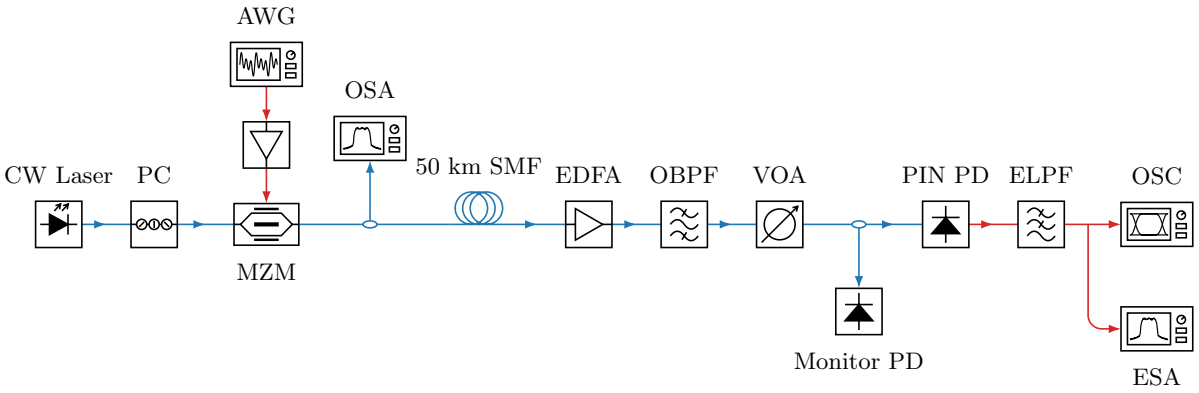


Labels: Component labels use a default clearance of `3pt`. Set `commtikz label distance=<dim>` on a picture, scope, or component, or use `label={[label distance=<dim>]above:Text}` for one label.
Visual Line Width Matching: Use `commtikz line width=<dimen>` (default `0.6pt`) or `commtikz match font` (`0.055em`) to tune stroke weight relative to the surrounding document font:

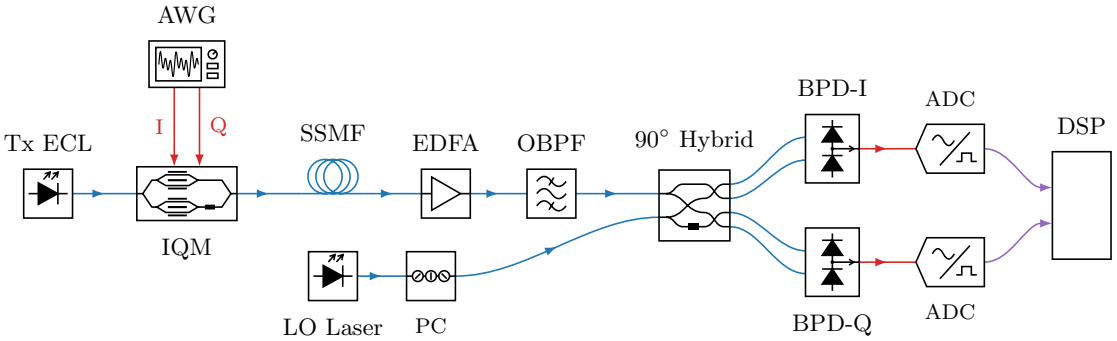


6 Example Electro-Optical & Circuit Schematics

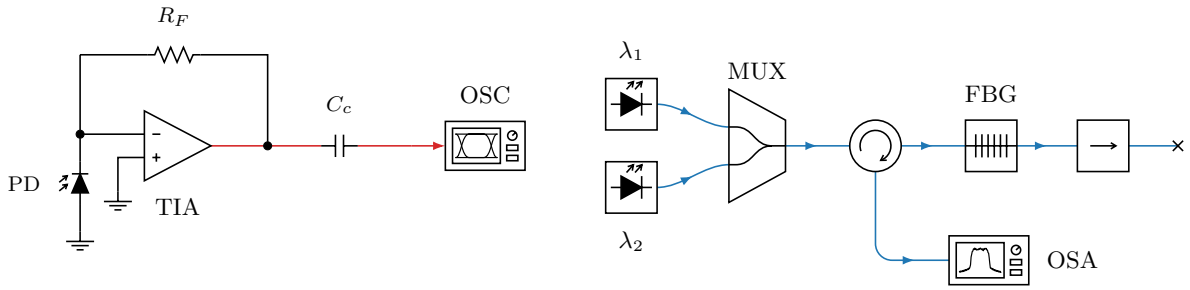
6.1 Setup 1: Intensity-Modulated Direct-Detection (IM/DD) Fiber Link



6.2 Setup 2: Coherent Optical DP-QAM Transceiver with 90° Hybrid & Balanced Detection



6.3 Setup 3: Photodiode TIA Front-End Circuit & WDM FBG Interrogation



Coupler colors: Standalone `coupler`/`couplercirc` and `fiberspool` nodes use the current drawing color (normally black). Inline `to[to coupler]` and `to[to couplercirc]` inherit the connector color; `to[to coupler={draw=black}]` overrides it.